

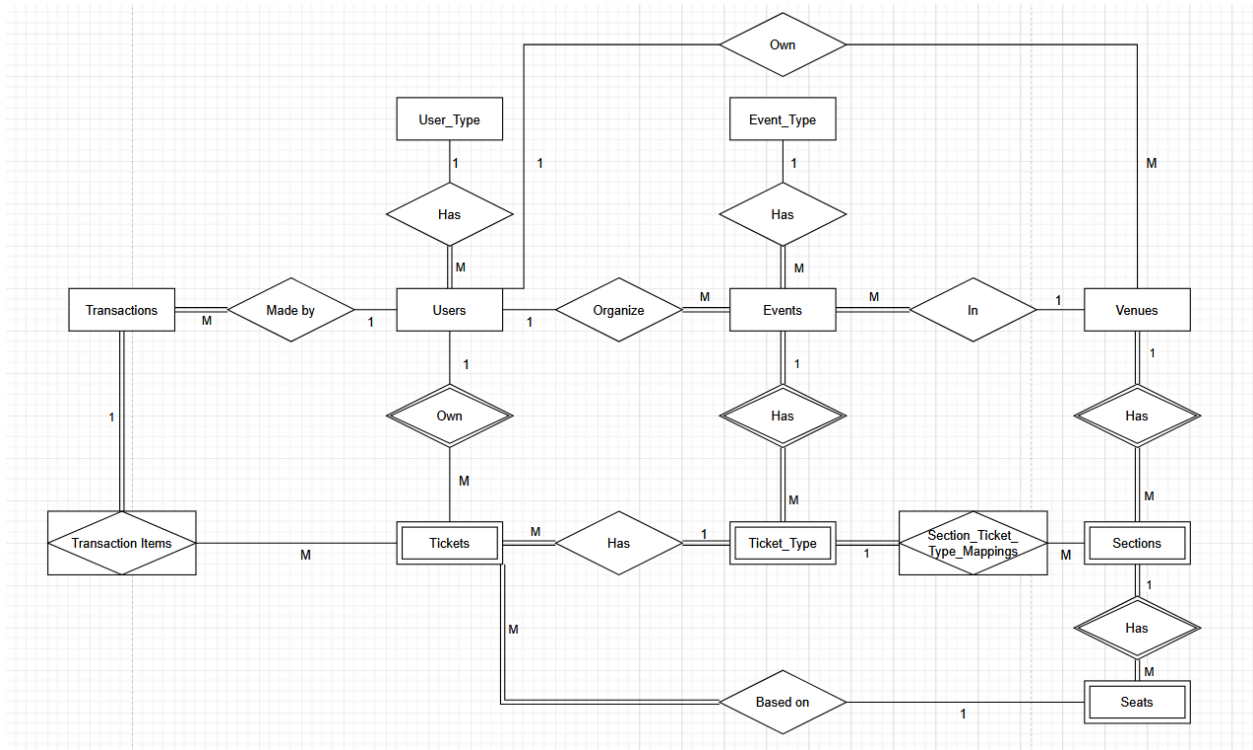
Database for Event Management and Ticketing Milestone 2

Yavuz Burak Yalçın
150220752

Alper Düzgün
150220312

Faruk Emre Karaca
150240723

ER Diagram



Tables

1. Events:

- event_id (primary_key, INT, auto increment)
- organizer_id (foreign_key, INT)
- venue_id (foreign_key, INT)
- event_type_id (foreign_key, INT)
- name (VARCHAR(100), Not NULL)
- date (DATE, Not NULL)

2. Event_Type:

- event_type_id (primary_key, INT)
- name (VARCHAR(100))

3. Users:

- user_id (primary_key, INT)
- type_id (foreign_key, INT)
- name (VARCHAR(100))
- Mail_address (VARCHAR(100))
- Phone_number (VARCHAR(20))

4. User_Type:

- user_type_id (primary_key)
- type_name (VARCHAR(100))
- authorization_level (INT)

5. Venues:

- venue_id (primary_key, INT)
- owner_id (foreign_key, INT)
- name (VARCHAR(100))
- country (VARCHAR(100))
- city (VARCHAR(100))
- address (VARCHAR(100))
- media_url VARCHAR(255)
- website_url (VARCHAR(100))
- seat_count (INT)
- section_count (INT)

6. Tickets:

- ticket_id (primary_key,INT)
- seat_id (foreign_key,INT)
- owner_id (foreign_key,INT, NULLABLE)
- ticket_type_id (foreign_key,INT)
- status (ENUM: available, reserved, sold)

7. Ticket_Type:

- ticket_type_id (primary_key,INT)
- event_id (foreign_key,INT)
- name (VARCHAR(100))
- price (FLOAT)

8. Transactions:

- transaction_id (primary_key,INT)
- user_id (foreign_key,INT)
- date (DATETIME)
- price (FLOAT)
- payment_method (VARCHAR(100))
- receipt_id (VARCHAR(100))
- installment_period (INT)

9. Transaction_Items:

- ticket_id (primary_key,foreign_key,INT)
- transaction_id (foreign_key,INT)

10. Sections:

- section_id (primary_key,INT)
- venue_id (foreign_key,INT)

- name (VARCHAR(100))

11. Seats:

- seat_id (primary_key,INT)
- section_id (foreign_key,INT)
- name (VARCHAR(100))

12. Section_Ticket_Type_Mapping:

- section_id (primary_key,foreign_key,INT)
- ticket_type_id (primary_key,foreign_key,INT)

Dummy Data for Some of the Tables

Table: Events										
event_id	organizer_id	venue_id	event_type_id	name		date				
Filter	Filter	Filter	Filter	Filter	Filter	Filter				
1 999	10	1	1	Tarkan Yaz Konseri		2024-07-20				

Table: Tickets										
ticket_id	seat_id	owner_id	ticket_type_id	status						
Filter	Filter	Filter	Filter	Filter						
1 1001	501	20	80	sold						
2 1002	502	NULL	80	available						
3 1003	601	NULL	81	available						

Table: Transactions										
transaction_id	user_id	date	price	payment_method	installment_period	receipt_id				
Filter	Filter	Filter	Filter	Filter	Filter	Filter				
1 50001	20	2024-05-10	1500.0	Credit Card		1 FTR1273961239GOV				

Table: Users										
user_id	type_id	name		mail_address	phone_number					
Filter	Filter	Filter	Filter	Filter	Filter					
1 10	1	Ahmet Organizatör		ahmet@biletix.com	555-0001					
2 20	2	Ayşe Müşteri		ayse@gmail.com	555-0002					

Table: Venues										
venue_id	owner_id	name		country	city	address	media_url	website_url	seat_count	section_count
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1 1	10	Harbiye Cemil Topuzlu		Turkey	Istanbul	Sisli	NULL	NULL	5000	2

Table: Transaction_Items		Filter in any column	
<u>ticket_id</u>	<u>transaction_id</u>		
Filter	Filter		
1	1001	50001	

Table: Section_Ticket_Type_Mapping		Filter in any column	
<u>section_id</u>	<u>ticket_type_id</u>		
Filter	Filter		
1	100	80	
2	101	81	

CRUD Operations

Create: New data is added when users register or when organizers set up venues and publish events. When a user buys a ticket, the system automatically creates a new record in the Transactions table to log the purchase.

Read: The system retrieves data to display information on the website. For example, the "Events" page reads data from the Events and Venues tables to show upcoming concerts. Users can also view their own purchase history by reading their records from the Transactions table.

Update: We use updates to keep information current. For instance, after a purchase, the system updates the Tickets table to change a ticket's status from "available" to "sold." Users can also update their personal profiles, such as changing their phone number or email address.

Delete: Delete operations are used mainly for cleanup by administrators. This allows admins to remove cancelled events from the system or delete user accounts if a user requests to leave the platform.